# Lab 1.1

Adomas Bieliūnas
2413914 22 var

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## 1.9

$$\sum_{k=20}^{30} \frac{i^k}{1+i} = \frac{1}{1+i} \sum_{k=20}^{30} i^k =$$

$$\frac{1}{1+i}(\cancel{i^{20} + i^{21} + i^{22} + i^{23}} + \cancel{i^{24} + i^{25} + i^{26} + i^{27}} + i^{28} + i^{29} + i^{30}) =$$

$$\frac{1}{1+i}(i^0 + i^1 + i^2) = \frac{1}{1+i}(1 + i + -1) = \frac{i}{1+i} =$$

$$\frac{0 \cdot 1 + 1 \cdot 1}{1^2 + 1^2} + i\frac{1 \cdot 1 - 0 \cdot 1}{1^2 + 1^2} = \frac{1}{2} + i\frac{1}{2}$$

## 2.9

$$|z| + \bar{u} + \frac{|w|}{w+1} = |i| + \overline{1+i} + \frac{|1-i|}{(1-i)+1} =$$

$$\sqrt{0^2 + 1^2} + 1 - i + \frac{\sqrt{1^2 + (-1)^2}}{2-i} = 1 + 1 - i + \frac{\sqrt{2}}{2-i} =$$

$$2 - i + \frac{\sqrt{2} \cdot 2 + 0 \cdot (-1)}{2^2 + (-1)^2} + i\frac{0 \cdot 2 - \sqrt{2} \cdot (-1)}{2^2 + (-1)^2} =$$

$$2 - i + \frac{2\sqrt{2}}{5} + i\frac{\sqrt{2}}{5} = \frac{2\sqrt{2} + 10}{5} + i\frac{\sqrt{2} - 5}{5}$$

## 3.9

$$z = -\frac{80}{3} - \frac{80i}{3}$$

$$r = |z| = \sqrt{(-\frac{80}{3})^2 + (-\frac{80}{3})^2} = \sqrt{\frac{2 \cdot 1600}{9}} = \frac{40\sqrt{2}}{3}$$

$$\theta = \arctan(\frac{-\frac{80}{3}}{-\frac{80}{3}}) - \pi = \arctan(1) - \pi = \frac{\pi}{4} - \pi = -\frac{3}{4}\pi$$

$$-\frac{80}{3} - \frac{80i}{3} = \frac{40\sqrt{2}}{3}e^{-i\frac{3}{4}\pi}$$

## 4.9

$$x^4 + 9x^2 + 20 = 0; u = x^2$$

$$u^2 + 9u + 20 = 0$$

$$\sqrt{D} = \sqrt{9^2 - 4 \cdot 1 \cdot 20} = 1$$

$$u_1 = \frac{-9+1}{2 \cdot 1} = -4$$

$$u_2 = \frac{-9-1}{2 \cdot 1} = -5$$

$$x_1^2 = -4 \Rightarrow x_{11} = 2i,\ x_{12} = -2i$$

$$x_2^2 = -5 \Rightarrow x_{21} = \sqrt{5}i,\ x_{21} = -\sqrt{5}i$$

## 5.9

$$n = 8, z = -1 - i$$

$$r = |z| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$\theta = \arctan(\frac{-1}{-1}) - \pi = \arctan(1) - \pi = \frac{\pi}{4} - \pi = -\frac{3}{4}\pi$$

$$z = \sqrt{2}e^{-i\frac{3}{4}\pi}$$

$$w_0 = \sqrt{2}^{\frac{1}{8}} e^{i\frac{-\frac{3}{4}\pi+2\pi\cdot 0}{8}} = \sqrt[16]{2}e^{-i\frac{3\pi}{32}} =$$

$$-\frac{2^{\frac{9}{16}}\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}}}{2} + \frac{2^{\frac{9}{16}}\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}}}{2} + \frac{2^{\frac{9}{16}}i\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}}}{2} + \frac{2^{\frac{9}{16}}i\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}}}{2},$$

$$w_1 = \sqrt{2}^{\frac{1}{8}} e^{i\frac{-\frac{3}{4}\pi+2\pi\cdot 1}{8}} = \sqrt[16]{2}e^{i\frac{5\pi}{32}} =$$

$$-\frac{2^{\frac{9}{16}}\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}}}{2} - \frac{2^{\frac{9}{16}}\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}}}{2} - \frac{2^{\frac{9}{16}}i\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}}}{2} + \frac{2^{\frac{9}{16}}i\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}}}{2},$$

$$w_2 = \sqrt{2}^{\frac{1}{8}} e^{i\frac{-\frac{3}{4}\pi+2\pi\cdot 2}{8}} = \sqrt[16]{2}e^{i\frac{13\pi}{32}} =$$

$$-\frac{2^{\frac{9}{16}}\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}}}{2} + \frac{2^{\frac{9}{16}}\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}}}{2} - \frac{2^{\frac{9}{16}}i\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}}}{2} - \frac{2^{\frac{9}{16}}i\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}}}{2},$$

$$w_3 = \sqrt{2}^{\frac{1}{8}} e^{i\frac{-\frac{3}{4}\pi+2\pi\cdot 3}{8}} = \sqrt[16]{2}e^{i\frac{21\pi}{32}} = -\sqrt[16]{2}\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}} + \sqrt[16]{2}i\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}},$$

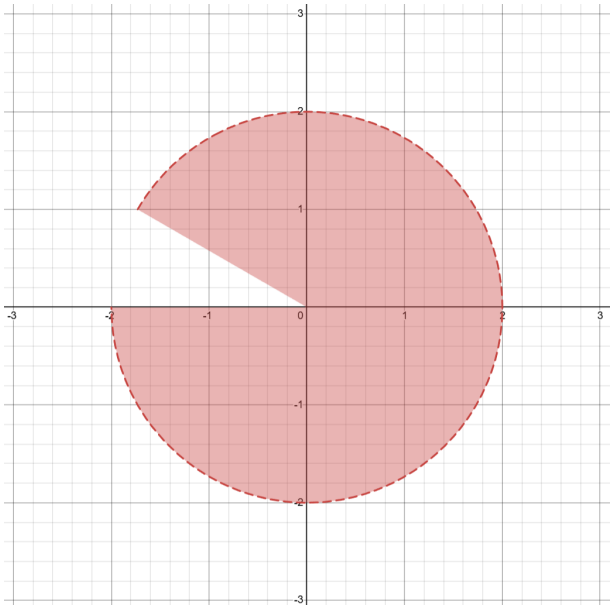
$$w_4 = \sqrt{2}^{\frac{1}{8}} e^{i\frac{-\frac{3}{4}\pi+2\pi\cdot 4}{8}} = \sqrt[16]{2}e^{i\frac{29\pi}{32}} = -\sqrt[16]{2}\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}} - \sqrt[16]{2}i\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}},$$

$$w_5 = \sqrt{2}^{\frac{1}{8}} e^{i\frac{-\frac{3}{4}\pi+2\pi\cdot 5}{8}} = \sqrt[16]{2}e^{-i\frac{27\pi}{32}} = \sqrt[16]{2}\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}} + \sqrt[16]{2}i\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}},$$

$$w_6 = \sqrt{2}^{\frac{1}{8}} e^{i\frac{-\frac{3}{4}\pi+2\pi\cdot 6}{8}} = \sqrt[16]{2}e^{-i\frac{19\pi}{32}} = \sqrt[16]{2}\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}} - \sqrt[16]{2}i\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}},$$

$$w_7 = \sqrt{2}^{\frac{1}{8}} e^{i\frac{-\frac{3}{4}\pi+2\pi\cdot 7}{8}} = \sqrt[16]{2}e^{-i\frac{11\pi}{32}} =$$

$$\frac{2^{\frac{9}{16}}\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}}}{2} + \frac{2^{\frac{9}{16}}\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}}}{2} - \frac{2^{\frac{9}{16}}i\sqrt{\frac{1}{2}-\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}}}{2} + \frac{2^{\frac{9}{16}}i\sqrt{\frac{\sqrt{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{1}{2}}}{2}+\frac{1}{2}}}{2}$$

## 6.9

$$|2z| \leq 4, Arg(\frac{z}{i}) < \frac{\pi}{3}$$
$$|2z| = 2\sqrt{x^2+y^2} \Rightarrow \sqrt{x^2+y^2} < 2 \Rightarrow x^2+y^2 < 4$$
$$Arg(\frac{z}{i}) = Arg(z) - \frac{\pi}{2} \Rightarrow Arg(z) < \frac{\pi}{3} + \frac{\pi}{2} \Rightarrow Arg(z) < \frac{5\pi}{6}$$
$$\sqrt{x^2+y^2} < 2$$
$$Arg(z) < \frac{5\pi}{6}$$